```
/* scanner for a toy Pascal-like language */

%{                              /* need this for the call to atof() below */
#include <math.h>
%}

DIGIT           [0-9]
ID              [a-z][a-z0-9]*

%%

{DIGIT}+        {
                printf( "An integer: %s (%d)\n", yytext,        // yytext is the current token
                atoi( yytext ) );
                }

{DIGIT}+"."{DIGIT}*
                {
                printf( "A float: %s (%g)\n", yytext,
                atof( yytext ) );
                }

if|then|begin|end|procedure|function
                { printf( "A keyword: %s\n", yytext ); }

{ID}            printf( "An identifier: %s\n", yytext );

"+"|"-"|"*"|"/" printf( "An operator: %s\n", yytext );

"{"[^}\n]*"}"           /* eat up one-line comments */

[ \t\n]+                /* eat up whitespace */
.                        printf( "Unrecognized character: %s\n", yytext );

%%
main( argc, argv )
int argc;
char **argv;
{
   ++argv, --argc; /* skip over program name */
   if ( argc > 0 )
   yyin = fopen( argv[0], "r" );
   else
    yyin = stdin;
   yylex();
}
```